

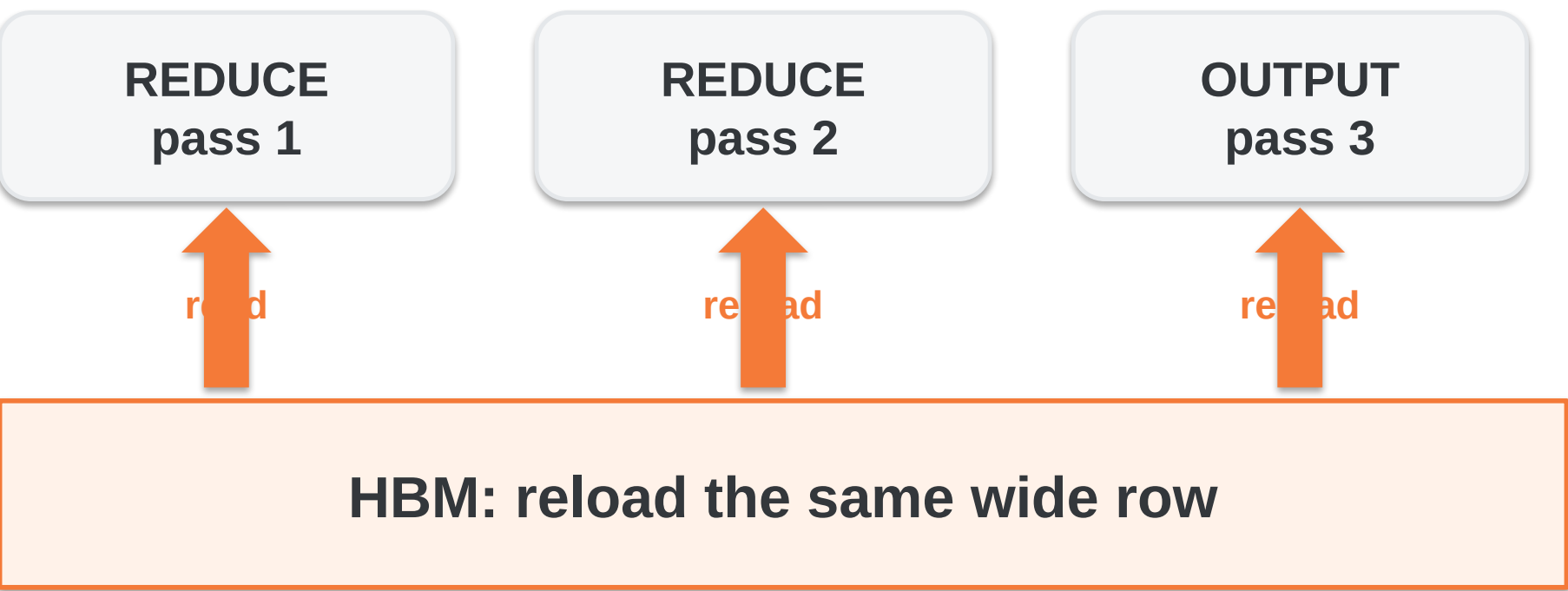
CREDIT: Cost-Guided DSMEM Tiling
for Wide Reduction-Reuse GPU Kernels

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1 THE PROBLEM

Wide rows hit the memory wall

Conventional multi-pass schedule



- One CTA cannot retain a wide row
- Later stages reread bulk data
- Naive clusters add barriers + remote traffic

DSMEM IS NOT FREE



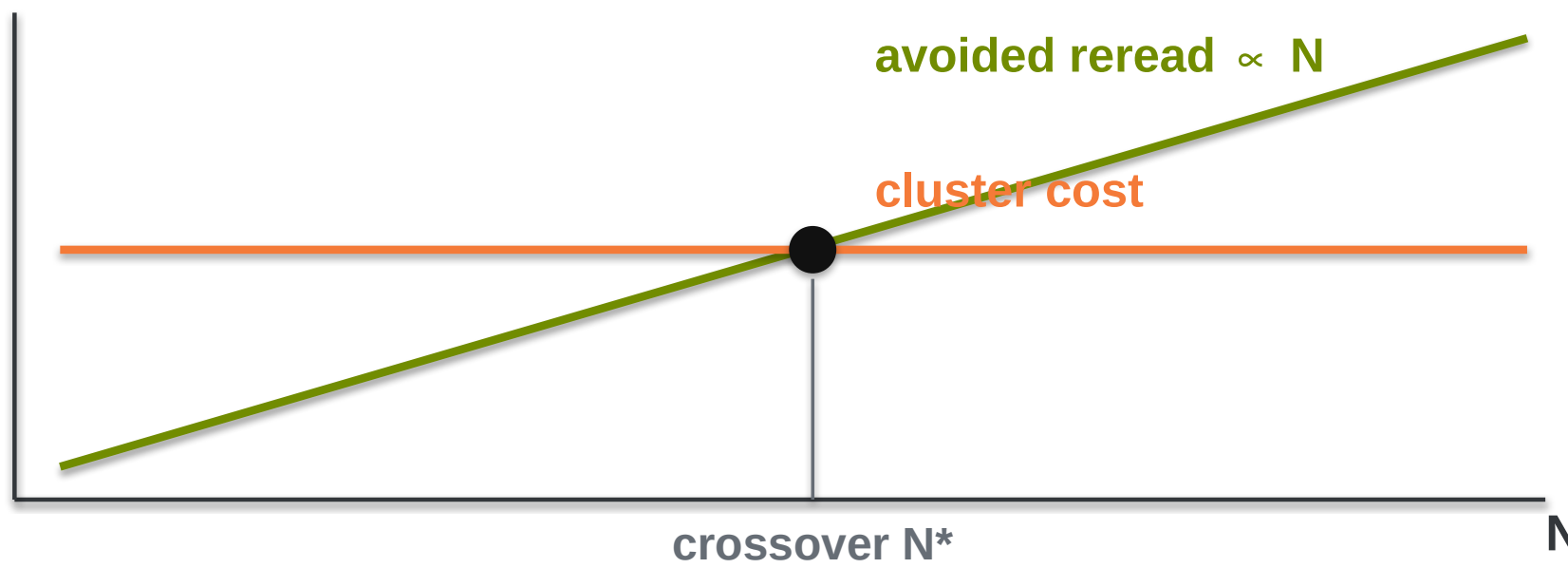
Cluster synchronization

404 cycles | RTX 5090

851 cycles | H100

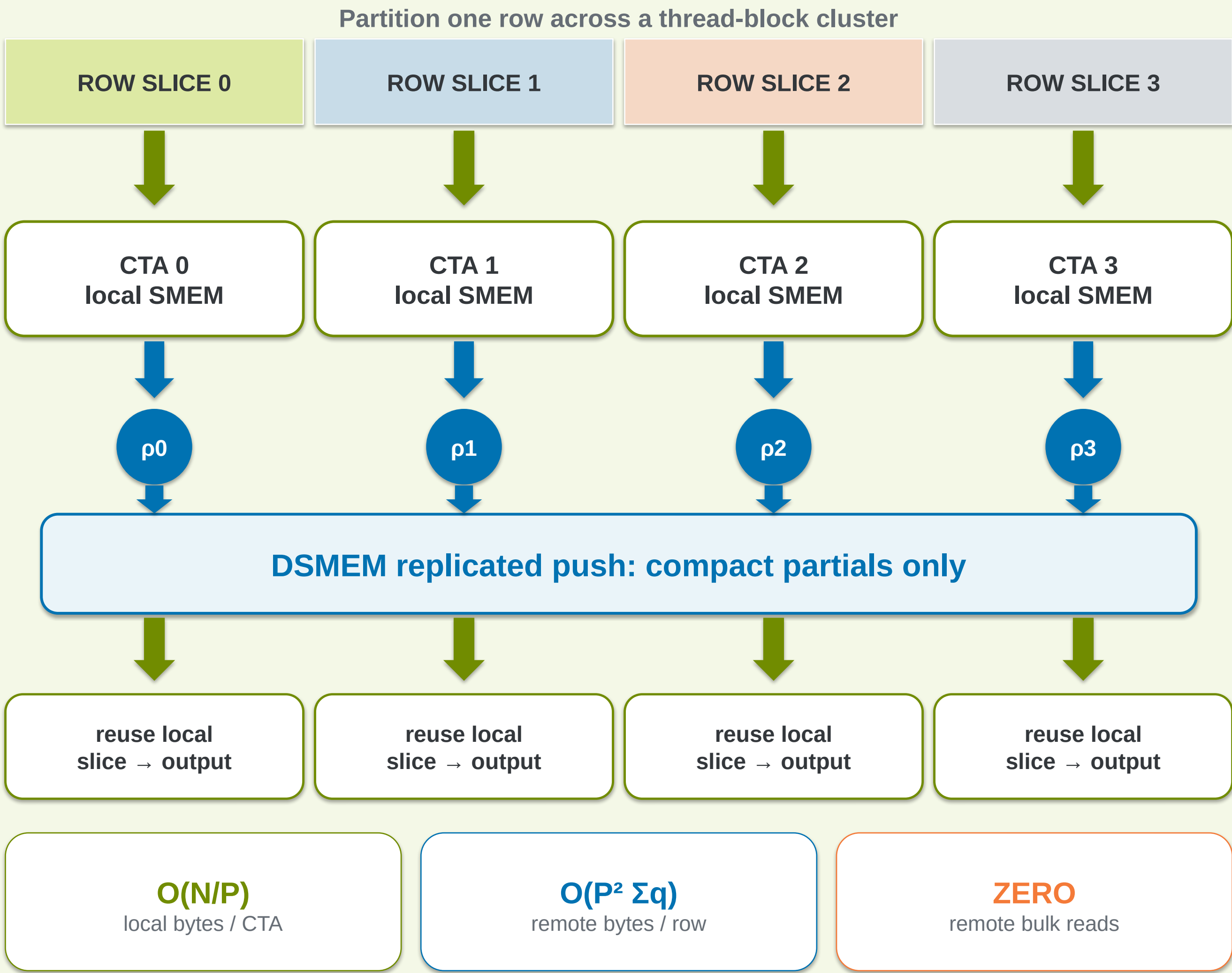
Can avoided rereads repay cluster overhead?

WHY WIDTH HELPS



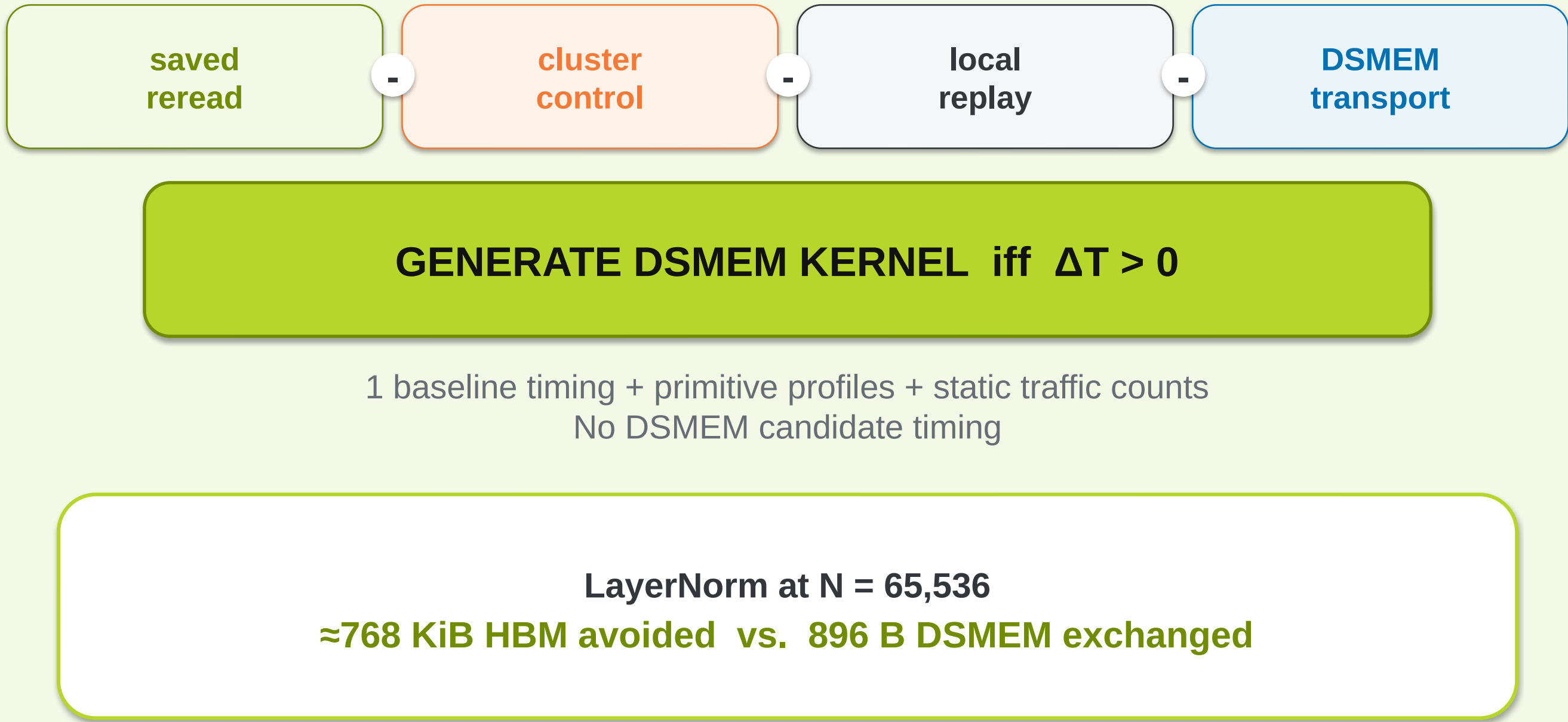
2 CREDIT

KEEP BULK DATA LOCAL
EXCHANGE ONLY SCALAR PARTIALS



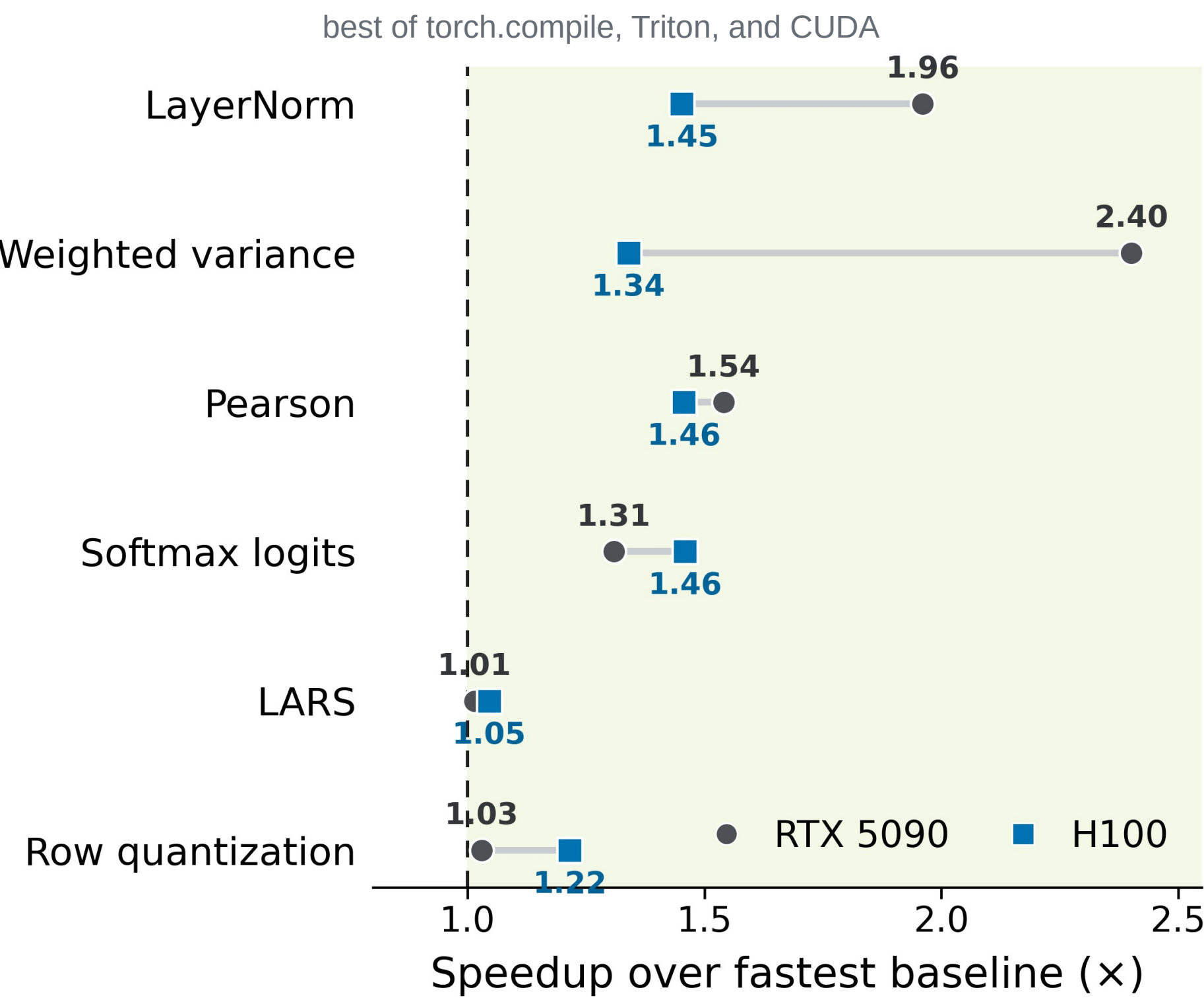
COST-GUIDED GENERATION

Predicted benefit =



3 CROSS-GPU EVIDENCE

ALL 6 WORKLOADS WIN AT 64K



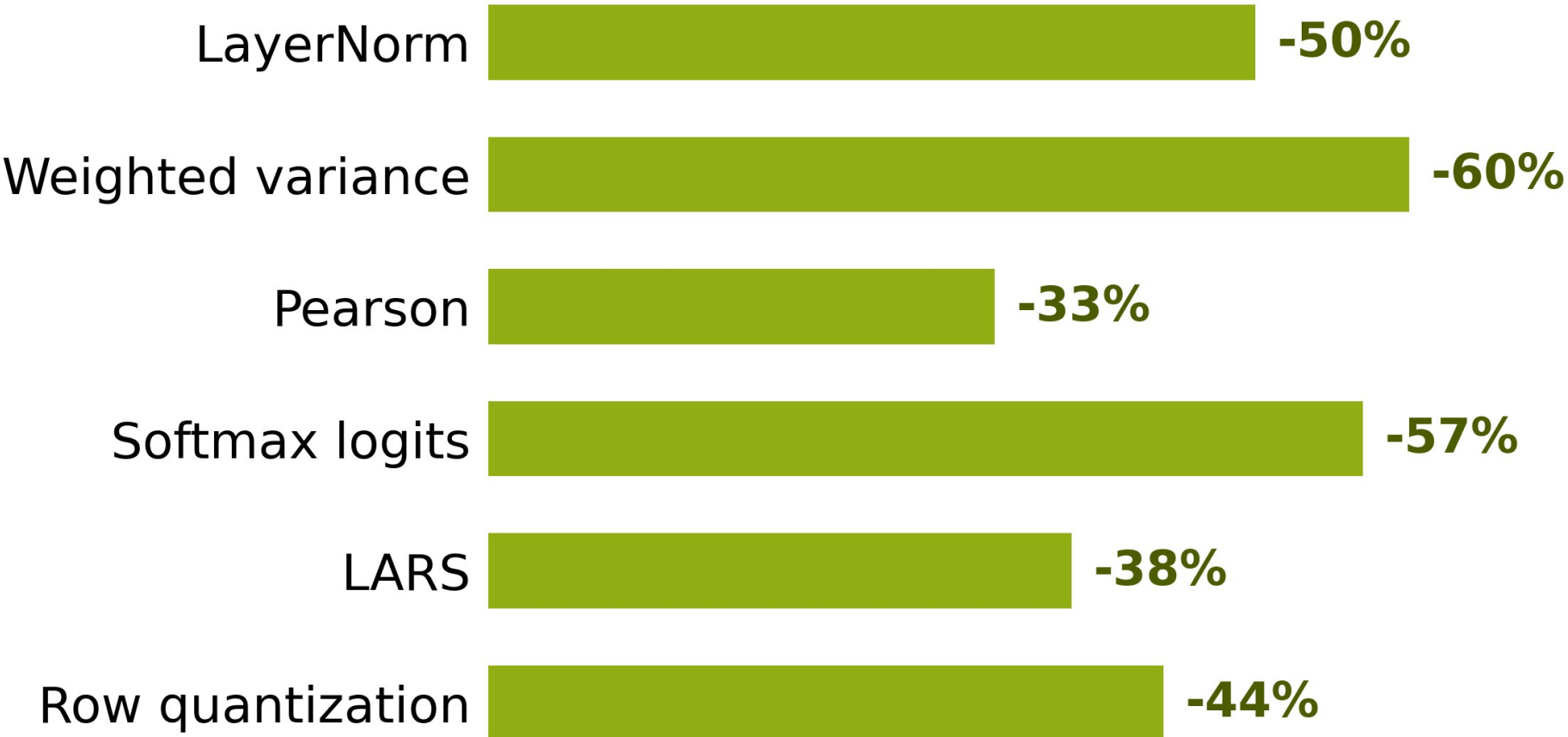
1.466x
RTX 5090 geomean

1.318x
H100 geomean

55 / 60
CUDA profit signs

MEASURED DRAM TRAFFIC ELIMINATED

RTX 5090, $N = 65,536$

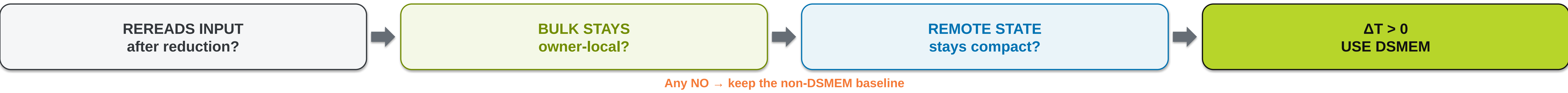


MODEL VALIDATION

90.0% RTX 5090 | 93.3% H100

Every crossover within one tested 2x width interval

4 DEPLOYMENT RULE



USE DSMEM

- Wide row + reduction + reuse
- Bulk slice stays owner-local
- Remote state stays compact

LayerNorm | softmax backward | pairwise statistics | quantization

AVOID DSMEM

- One-pass reduction
- Scan / selection
- Bulk halo exchange

Width amortizes overhead; width alone is not predictive.

Wide rows amortize overhead; width alone is not a selection rule.

DSMEM is a selective reduction-reuse mechanism, not a general fusion fabric.